

Danko Lobos Bustamante

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[linkedin.com/in/dankollb/](https://www.linkedin.com/in/dankollb/) – Santiago, Metropolitan Region, Chile.

Computer Science Engineer

Recently graduated. Experience in software engineering, artificial intelligence, and IT support. Worked on optimizing AI models, developing full-stack applications and data science research projects. Applied software engineering methodologies and best practices to real-world projects. For my thesis, I implemented a specialized full-stack web application for managing, analyzing, and visualizing genomic data.

Education

Metropolitan University of Technology – Ñuñoa campus, Santiago, Chile.

Professional Degree in Computer Science Engineering – Obtained in March 2026.

- Teaching Assistant in the Computer Science Department for eight consecutive semesters..
- Research Assistant in Animal-Computer Interactions (publication in ACI '22).

University of Chile – Beauchef campus, Santiago, Chile.

Bachelor of Science in Computer Science Engineering – Enrolled from 2018 to 2021.

- Vice President of the Computer Science Students Union (CADCC), 2020.

Work Experience

Delirium.AI – Santiago, Chile.

AI Engineer – Internship – October 2024 to December 2024.

Objective: To collaborate in the development of a web application for monitoring using computer vision AI applied to industrial safety.

- **Data management and preprocessing:** Expanded training dataset through ETL processes on large image datasets using specialized software. Built automated Python data pipelines to optimize image collection and validation, significantly reducing manual effort and improving dataset consistency.
- **Model development, testing and optimization:** Evaluated model performance using various metrics, fine-tuned hyperparameters, and debugged issues to ensure accuracy, efficiency, and reliability. Improved object detection accuracy by 20% through iterative model optimization and data augmentation techniques.
- **Integration and deployment:** Designed and implemented a new database and RESTful API with multiple endpoints from scratch. Developed web application functionality to visualize AI model outputs and performance data, enabling real-time insights and improved user interaction.

Alloxentric – Santiago, Chile.

Software Engineer – Traineeship Agreement – August 2024 to December 2024

- **Full-stack web application development:** worked on the development of a web application for human resources recruitment processes, including modules for creating and tracking vacancies and candidates. Implemented new frontend and backend functionalities, including forms, validations, and business logic aligned with the client's requirements.
- **RESTful API design and integration:** designed and integrated RESTful services and APIs for managing user data and selection processes, ensuring data consistency and traceability.
- **Agile project management:** applied the Earned Value Management (EVM) methodology to monitor project progress, deadlines, and performance throughout the semester. Worked in teams in an agile environment, participating in planning, monitoring, and continuous improvement activities for the solution.

DIRECTIC Tech Solutions – Santiago, Chile.

IT Support Engineer – Independent Contractor Agreement – January 2022 to March 2024.

- **Client communication:** Provided technical support and troubleshooting for 1,000+ users across 100+ companies in the Chilean mining industry through online and remote assistance.
- **Database management:** Performed database backups, data recovery and data imports/exports.
- **System and network monitoring:** Monitored the performance and maintained their configurations.
- **Quality assurance:** Executed and documented QA tests to ensure software reliability and performance.
- **Documentation:** Wrote and maintained technical documentation, including issue reports, resolutions, and standard procedures.

Latin American Women in Tech (LATINITY) – Santiago, Chile.

Web Developer – Independent Contractor Agreement – June 2020 to July 2020.

- Implemented a new minimalist website for the LATINITY conferences
- Managed the hosting service and uploaded the information from the organization and past versions of the conference.
- Live site: <https://latinity.info/>

Technologies

Main:

- **Programming Languages:** Python, Java, JavaScript, C, C++.
- **IA & Data Science:** PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, ChromaDB, Ollama.
- **Web Development:** Angular, Django, Vue, HTML, CSS/SCSS, TypeScript, XML, XSLT.
- **Database Management:** PostgreSQL, MySQL, MS SQL Server, MongoDB.

Others:

- **Cloud Computing:** AWS, GCP, IBM Cloud.
- **DevOps & Tools:** Docker, Git, Postman, Keycloak.
- **Agent Automation:** n8n, Dify, MCP.
- **Other Languages:** C#, PHP, R.
- **Operating Systems:** Windows, Linux (Debian, RHEL, Fedora), macOS.

Certified Courses

- *Project Management and Agile Methodology Fundamentals* – Santander Open Academy (2025).
- *Introduction to Cloud* – AWS Academy (2024).

Methodologies & Best Practices

- Earned Value Management Methodology (EVM).
- Agile Project Development and Management (Agile).
- Test Driven Development (TDD).
- Software Design Patterns (SDP).

Languages

- Spanish: Native.
- English: B2 Level.

Research

1. Jérémy Barbay, Daniel Freire-Fernández, and Danko Lobos-Bustamante. 2023. *Popping Up Balloons for Science: a Research Proposal*. In Proceedings of the Ninth International Conference on Animal-Computer Interaction (ACI '22). Association for Computing Machinery, New York, NY, USA, Article 12, 1–5. <https://doi.org/10.1145/3565995.3566029>.